Appendix A

Analysis

A.1 Sets

We denote sets with curly brackets. In the simplest case, the elements of a set are enumerated inside those brackets, e.g.

$$\{a, b, c\}$$

is a set that contains the elements a, b and c (whatever those might be), and nothing else. Often it is convenient to give sets (or other objects) labels or names, in order to avoid having to give the definition over and over again. For example, we may wish to give the set which we have just defined the name A. The basic symbol of a definition is $:=$, where the object to be defined is on the left. So we write

$$A := \{a, b, c\}.$$

The notation of belonging is the usual $a \in A$. (The negation, i.e. "does not belong" is denoted by the symbol $\notin$.) Moreover, if A and B are sets, we write $A \subset B$ if all the elements of A are also elements of B. We say A is a subset of B. The set of all elements that are both in A and B is called the *intersection* of the two sets, written as $A \cap B$. The set of all elements that are either in A *or* in B is called the union of the sets, written $A \cup B$.

Often, it is not possible to enumerate all elements belonging to a set explicitly. For example, if we denote the set of all natural numbers as $\mathbb{N}$ we might write

$$\mathbb{N} := \{1, 2, 3, \dots\}.$$

We use this particular notation (i.e. the dots $\dots$) only if it appears clear how to complete the set. Alternatively, we often use a colon to separate a variable that stands for the elements of a set from a statement that (hopefully) uniquely specifies the elements. For example, we may wish to define the set X containing all integer numbers greater than 5. We could write

$$X := \{n : n \in \mathbb{N},\, n > 5\}, \tag{A.1}$$

where the "statement" reads: "n is an integer number and n is greater than 5". Note that we avoid using mathematical symbols for "and" in this statement. We simply use a comma or semicolon to separate different parts of the statement. For better or worse, this is quite the usual way to do it.

Sometimes we may wish to *pair* elements of two (or more) sets. A particular way of creating such a pairing is the Cartesian product of two sets

$$X \times Y := \{(x, y) : x \in X, y \in Y\}. \tag{A.2}$$

We have thus created a set of *ordered pairs* (x, y), where "ordered" simply means that the first element of the pair always comes from the set X, and the second element comes from the set Y. The sets can be the same. An example is the definition of the two-dimensional Euclidean vector space $\mathbb{R}^2 := \mathbb{R} \times \mathbb{R}$ as the Cartesian product of the real line. Cartesian products can be extended to more than two sets in an obvious manner.

A.2 Functions

If we see a statement like $f(x) = x + 1$, we immediately interpret this as a mapping between the real numbers, that maps any $x \in \mathbb{R}$ to $x + 1$. A more formal way of writing such a function is $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x + 1$. In general we say

$$f : X \to Y \tag{A.3}$$

is a mapping that assigns each member of the set X exactly one member of the set Y. The value of the function that is assumed under the mapping at $x \in X$ is then called $f(x) \in Y$. Note that our basic notation of a function first identifies the sets between which the mapping takes place. This may seem redundant in simple cases. However, in this course we define mappings between a variety of different sets (other than the real numbers). So the notation will come in handy.

A.2.1 Real-Valued Functions

We call a function that maps into (a subset of) $\mathbb{R}$ a *scalar* function[1], and a function that maps into (a subset of) $\mathbb{R}^d$, for $d > 1$, a vector-valued function. We shall often use **boldface** notation to identify vectors and vector-valued functions.

Let $\mathcal{D} \subset \mathbb{R}$, and consider a function $y : \mathcal{D} \to \mathbb{R}$. The *derivative* is defined

$$y'(x_0) := \lim_{h \to 0} \frac{y(x_0 + h) - y(x_0)}{h}, \tag{A.4}$$

provided the limits exist, in which case y is said to be *differentiable* at $x_0 \in \mathcal{D}$. Usually, whenever one speaks of a differentiable function, one means *continuously differentiable*, i.e. a unique limit in (A.4) must be approached for any sequence $\{h_n\}_{n \in \mathbb{N}}$ with $x + h_n \in \mathcal{D}$ and $\lim_{n \to \infty} h_n = 0$. If $\mathcal{D}$ is a closed interval, however, then derivatives at the boundaries are well-defined only "from one side", using $h > 0$, or $h < 0$, as appropriate.

Recursive definition yields higher derivatives $y'', y''' \ldots$ ("second", "third" derivative, etc). Sometimes we prefer to denote higher derivatives with integer superscript, e.g. $y^{(k)}$ is the kth derivative.

Let $\mathcal{D} \subset \mathbb{R}^d$ with $d > 1$, and let $\mathbf{x} = (x_1, \ldots, x_d)^T \in \mathcal{D}$. Consider a function $u : \mathcal{D} \to \mathbb{R}$. The classic definition of a partial derivative is

$$\frac{\partial u}{\partial x_j}(\mathbf{x}_0) := \lim_{h \to 0} \frac{u(\mathbf{x}_0 + h\mathbf{e}_j) - u(\mathbf{x}_0)}{h},$$

where $\mathbf{e}_j$ is the jth Cartesian unit vector $\mathbf{e}_j = (0, \ldots, 1, \ldots, 0)^T$, i.e. the "1" is in the jth place. If the limit exists, we say $u(\mathbf{x})$ possesses the *partial derivative* with respect to x_j at $\mathbf{x}_0 \in \mathcal{D}$. Recursively, we define higher partial derivatives, e.g. a second partial derivative

$$\frac{\partial^2 u}{\partial x_j \partial x_k}(\mathbf{x}_0) := \lim_{h \to 0} \left\{ \frac{1}{h} \left(\frac{\partial u}{\partial x_j}(\mathbf{x}_0 + h\mathbf{e}_k) - \frac{\partial u}{\partial x_j}(\mathbf{x}_0) \right) \right\}.$$

Often we shall write

$$u_{x_j} := \frac{\partial u}{\partial x_j}, \qquad u_{x_j x_k} := \frac{\partial^2 u}{\partial x_j \partial x_k}.$$

Also we sometimes use more descriptive symbols for the variables. If one coordinate represents time, we use the symbol t, and for spatial two- or three-dimensional coordinates we sometimes use (x, y) and (x, y, z), respectively.

It is obvious how higher partial derivatives can lead to complicated expressions. To avoid this, one often uses *multi-index* notation. Let

$$\alpha := (\alpha_1, \ldots, \alpha_d),$$

[1] We could also consider mappings between complex numbers using rather similar notation. For simplicity we stick to real numbers in this section.

where each α_j is a non-negative integer. The vector α is called a multi-index of order

$$|\alpha| := \alpha_1 + \cdots + \alpha_d.$$

Then we define

$$D^\alpha u(x) := \frac{\partial^{|\alpha|} u(x)}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}.$$

In particular, we define the set of derivatives of order k as

$$D^k u(x) := \{D^\alpha u(x) : \ |\alpha| = k\}$$

If we assign a particular ordering to such a set, we may define a tuple $(D^{\alpha_1} u(x), \dots, D^{\alpha_{n_k}} u(x))$, where n_k depends on the order k of differentiation, and interpret the above set of all partial derivatives of order k as a point in $\mathbb{R}^{n_k}$. This notation will be convenient for writing PDE in a general way.

The special case of mappings between real (or complex) numbers (or, more generally, vectors thereof) are so common, we shall reserve the term "function" for this case. For mappings between general X and Y, we stick with the generic term, and say f defines a "mapping" or say that f is an "operator".

Yet another special case of nomenclature is usually reserved for the case that a function is mapped to the real (or complex) numbers. Such mappings $l : V \to \mathbb{R}$ are often called functionals. Note that this is, in general, different from a simple function between the real numbers, as the set V can be more general.

A.2.2 Order Symbols

We often need to estimate the magnitude of a function by comparing it to another function. Order symbols are merely a convenient notation for this. We give a formal definition for functions defined on the real numbers. There are rather obvious extensions, for example for sequences. Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions, and let $x_0 \in \mathbb{R}$. We say

$$f(x) = \mathcal{O}(g(x)) \qquad (\text{as } x \to x_0), \tag{A.5}$$

(read: "f is of order g, as x goes to x_0") if there are positive constants C, δ, such that

$$|f(x)| \leq C|g(x)|, \qquad \text{whenever } |x - x_0| < \delta, \tag{A.6}$$

We say that f is of order g as x goes to infinity, if there exists a positive constant C and $\overline{x} \in \mathbb{R}$, such that

$$|f(x)| \leq C|g(x)|, \qquad \text{for } x > \overline{x}. \tag{A.7}$$

Note that certain limits (e.g. $x \to x_0$) are involved in the preceding definitions. Often it is clear from the context which limit is to be taken, and we simply say "f is of order g", without stating or writing the limit explicitly. Also note that, in the statement $f(x) = \mathcal{O}(g(x))$ in (A.5), the constant C in the definition (A.6) or (A.7) is not included. This is because usually the function g by which we can bound the function f is of far greater importance than the constant involved.